

A Multimodal Approach for Detecting Deepfakes in Video and Audio

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Abstract—Synthetic media generated through deep learning has reached a level of perceptual realism that makes manual verification increasingly unreliable. Such content, broadly termed deepfakes, introduces substantial risks in areas spanning misinformation, identity fraud, and digital forensics. Single-modality detection systems that examine either video frames or audio signals in isolation remain vulnerable to high-quality manipulations that are specifically crafted to exploit per-modality weaknesses. This paper presents a joint audio-visual deepfake detection framework that processes both streams independently and combines their predictions through a late-fusion strategy. The visual stream is handled by a transfer-learned Xception convolutional backbone coupled with a Bidirectional Gated Recurrent Unit that captures temporal inconsistencies across a sequence of 24 facial crops per video. The acoustic stream is handled by a four-block Depthwise-Separable Convolutional Network operating on 128-band Mel spectrograms augmented with time and frequency masking. Evaluated on FaceForensics++ (video) and the Fake-or-Real corpus (audio), the system achieves 76.67% accuracy with macro F1 of 0.77 on the video task and 98.00% accuracy with macro F1 of 0.98 on the audio task. A Streamlit-based web application provides real-time inference with graceful fallback when either modality is absent.

Index Terms—Deepfake Detection, Multimodal Fusion, Xception, Bidirectional GRU, Depthwise-Separable CNN, SpecAugment, Mel Spectrogram, FaceForensics++, Fake-or-Real Dataset, Audio-Visual Analysis

I. INTRODUCTION

Advances in generative deep learning have made it possible to fabricate audio-visual content that is perceptually indistinguishable from genuine recordings [1], [2]. Face-swapping pipelines, voice cloning systems, and lip-synchronization tools are now widely accessible, enabling the creation of videos in which a target person appears to speak or act in ways they never did. Collectively referred to as deepfakes, such fabrications have been used to spread political disinformation, produce non-consensual intimate imagery, and enable identity-based financial fraud [3].

Exploiting those signatures is not straightforward. Generation pipelines change rapidly across tool versions, and common post-processing steps such as re-encoding and resolution downscaling tend to destroy fragile low-level cues. Adding to this challenge, real-world media is frequently monomodal: a video clip may arrive without an audio track, or an audio recording may lack any accompanying imagery. A practical detection system therefore needs to exploit whatever modal-

ities happen to be present while still producing a reliable confidence estimate [22].

We designed our work around these constraints. The contributions of this paper are as follows:

- A two-branch detection framework in which an Xception-BiGRU video model and a four-block Separable-CNN audio model are trained independently on domain-appropriate benchmarks.
- Application of SpecAugment time-frequency masking to audio spectrograms, effectively doubling the training corpus and improving spectral robustness.
- A late-fusion inference module with modality-level fallback, enabling deployment on incomplete multimedia inputs.
- Detailed per-class evaluation including precision, recall, F1-score, and confusion matrices, alongside a comparative analysis with published methods on the same benchmarks.

II. RELATED WORK

A. Early Artifact-Based Video Detection

Work on video deepfake detection originally focused on identifying generation artifacts that were characteristic of early face-swap models. Li and Lyu [5] observed that GAN-synthesized faces frequently failed to produce coherent eye blinking due to the scarcity of such frames in training data, and exploited this physiological constraint as a binary detector. Matern et al. [6] extended this line of reasoning to a broader set of facial attribute anomalies including irregular skin colour, missing specular reflections, and asymmetric pupil shapes. These handcrafted features worked well against first-generation tools; however, as generators grew more powerful, they became less and less effective.

B. Deep Learning Approaches to Video Forgery

Afchar et al. [7] proposed MesoNet, a compact CNN designed to capture mesoscopic image statistics rather than high-level semantic features. The architecture generalized across manipulation types while remaining computationally inexpensive. The FaceForensics++ benchmark introduced by Rössler et al. [8] unified evaluation across four manipulation methods—Deepfakes, Face2Face, FaceSwap, and NeuralTextures—and confirmed that CNN-based detectors

substantially outperform human observers even under heavy compression. In that benchmark, Chollet’s Xception architecture [9] emerged as the strongest single-frame detector, with its depthwise-separable convolutions proving particularly effective at capturing fine-grained texture artifacts.

Temporal modeling was subsequently introduced to detect inconsistencies spanning multiple frames rather than individual ones. Sabir et al. [10] paired a CNN feature extractor with a recurrent network to flag frame-level discontinuities arising from imperfect temporal blending. Tariq et al. [11] built a Convolutional LSTM residual architecture in which skip connections stabilize gradient flow during temporal back-propagation, yielding resilience to common video post-processing. Jamsheed and Bhanu [12] subsequently showed that processing CNN features bidirectionally outperforms unidirectional baselines, since the model can draw on future context when evaluating any given frame.

Transformer-based detectors have more recently attracted interest. Wodajo and Atnafu [13] showed that Vision Transformers applied to face crops achieve competitive accuracy with fewer inductive biases than CNNs. Wang et al. [14] proposed FakeSpotter, a neuron-coverage approach that monitors activation distributions of a pretrained network rather than relying on explicit visual artifacts.

C. Audio Deepfake Detection

The ASVspoof challenge series [15] has been the primary driver of anti-spoofing research, providing standardized corpora and evaluation protocols. Classical methods rely on handcrafted features such as MFCCs, Linear Predictive Coding coefficients, and CQCCs paired with Gaussian Mixture Models. These were quickly superseded once deep learning was applied; Alzantot et al. [16] demonstrated that a lightweight CNN operating on raw waveforms outperformed feature-engineering pipelines on standard spoofing benchmarks. The Fake-or-Real corpus [17] offers balanced real and GAN-synthesized speech samples recorded under diverse acoustic conditions, making it well-suited for training robust audio classifiers. Gong et al. [19] showed that a pure transformer architecture operating on Mel spectrograms—the Audio Spectrogram Transformer—achieves state-of-the-art results across multiple audio classification tasks, reinforcing the value of spectrogram-based representations.

D. Multimodal and Cross-Modal Detection

Zhou et al. [21] argued that audio-visual synchrony is a reliable forgery cue because face-swap tools typically alter only the visual stream, creating temporal mismatches between lip movement and speech acoustics. Mittal et al. [22] extended this idea by using emotional consistency between spoken content and facial expressions as an auxiliary training signal. Kim et al. [23] proposed FReTAL, which transfers discriminative knowledge from a teacher to a student network through representation distillation, enabling cross-dataset generalization. Alhaji et al. [24] further showed that combining ACO-PSO meta-heuristic feature selection with deep CNN classification

reduces dimensionality while preserving discriminative power across multiple benchmarks.

III. METHODOLOGY

A. System Overview

The detection framework consists of two independently trained branches and a shared inference-time fusion module. The video branch accepts raw video clips and outputs a two-class probability vector over {Real, Fake}. The audio branch accepts audio waveforms and outputs an independent probability vector over the same classes. During inference the two vectors are averaged element-wise, yielding a fused posterior from which the final label is derived by taking the argmax. Fig. 1 presents the complete end-to-end architecture.

B. Datasets and Partitioning

Video – FaceForensics++. The benchmark contains compressed video sequences spanning four manipulation categories: Deepfakes, Face2Face, FaceSwap, and NeuralTextures [8]. For this study, 400 clips are used—200 genuine recordings and 200 fabricated clips drawn equally from the four manipulation types. All clips are processed at their H.264-compressed quality level (compression factor 23). A stratified 70/15/15 split yields 280 training, 60 validation, and 60 test clips with balanced class ratios in each partition.

Audio – Fake-or-Real (FoR). The normalized FoR corpus [17] contains recordings of genuine speech alongside GAN-synthesized utterances generated by multiple text-to-speech and voice-conversion systems. We use 1,000 utterances—500 real and 500 fake—applying the same 70/15/15 stratified split to obtain 700 training, 150 validation, and 150 test samples. Both classes are equally represented in every partition, so class weights are set to unity.

TABLE I
DATASET PARTITIONS USED IN THIS WORK

Branch	Corpus	Train	Val	Test
Video	FaceForensics++ (200R/200F)	280	60	60
Audio	Fake-or-Real (500R/500F)	700	150	150

C. Video Branch: Xception-BiGRU

Frame extraction and face detection. Each video is decoded frame-by-frame using OpenCV. The MTCNN detector [20] locates the face bounding box in each frame; if detection fails the full frame is downsampled and used as a substitute. Exactly $N=24$ frames are selected by uniform temporal sampling, ensuring coverage of the complete video duration regardless of clip length. Each face crop is resized to 128×128 pixels in RGB and normalized to the unit interval. Optional augmentation during training applies horizontal mirroring, rotations drawn from $\mathcal{U}(-10^\circ, +10^\circ)$, and small brightness jitter.

Spatial representation. Every frame in the 24-frame sequence is encoded by an Xception network [9] pretrained on ImageNet. To process the temporal dimension without nested

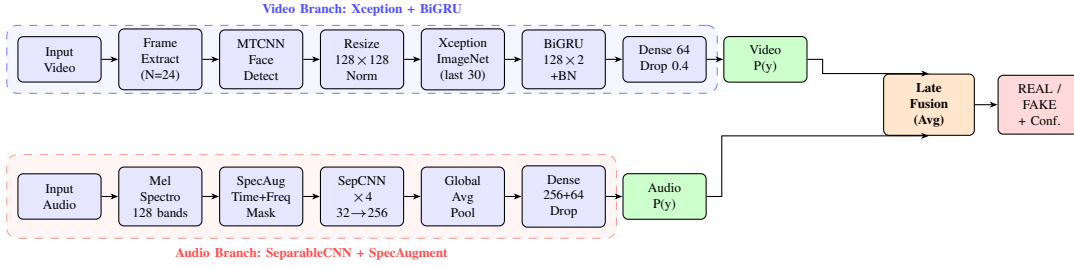


Fig. 1. End-to-end architecture of the proposed multimodal deepfake detection system. The two branches operate independently during both training and deployment. At inference time, the softmax outputs of both branches are averaged to produce a fused prediction.

TimeDistributed overhead, the batch and time axes are merged: a tensor of shape $(B, 24, 128, 128, 3)$ is reshaped to $(24B, 128, 128, 3)$, forwarded through Xception with global average pooling to produce a 2048-dimensional embedding per frame, then rearranged back to $(B, 24, 2048)$. The depthwise-separable convolution at each Xception layer computes:

$$(I * K)(i, j) = \sum_m \sum_n I(m, n) K(i - m, j - n), \quad (1)$$

where channel-wise and point-wise filters are applied sequentially, greatly reducing the parameter count compared to standard convolution at equivalent receptive fields. The final 30 layers of Xception are kept trainable; all preceding layers are frozen to preserve low-level ImageNet features while permitting fine-tuning of high-level representations.

Temporal modeling. A Batch Normalization layer stabilizes the 24-step feature sequence before it enters a Bidirectional GRU (BiGRU) with 128 units per direction. In each direction the GRU governs information flow through two learned gates:

$$z_t = \sigma(W_z[h_{t-1}, x_t] + b_z), \quad (2)$$

$$r_t = \sigma(W_r[h_{t-1}, x_t] + b_r), \quad (3)$$

$$\tilde{h}_t = \tanh(W_h[r_t \odot h_{t-1}, x_t] + b_h), \quad (4)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t. \quad (5)$$

The update gate z_t controls how much past memory is carried forward, while the reset gate r_t determines how much context influences the candidate state $\tilde{h}_t$. Forward and backward final hidden states are concatenated to form a 256-dimensional summary that encodes both causal and anti-causal temporal context—enabling the detection of artifacts such as inter-frame flickering, non-monotonic motion trajectories, and abnormal blink cadences that span multiple frames. L2 regularization is applied to kernel ($\lambda_k=0.005$) and recurrent ($\lambda_r=0.001$) weight matrices.

Classification head. The BiGRU output passes through Dropout(0.50), a Dense layer with 64 units and ReLU activation, Dropout(0.40), and a final Softmax(2) layer. Total trainable parameters (excluding frozen Xception layers): 1,693,378.

D. Audio Branch: Separable-CNN with SpecAugment

Preprocessing. Each audio recording is converted to single-channel format and resampled to 16 kHz. Files shorter than

5 s are zero-padded at the tail; files longer than 5 s are centre-cropped. A 128-band Mel spectrogram is derived from the Short-Time Fourier Transform (STFT):

$$\text{STFT}\{x[n]\}(m, \omega) = \sum_{n=-\infty}^{\infty} x[n] w[n - m] e^{-j\omega n}, \quad (6)$$

where $w[\cdot]$ is a Hann window of length 512 with 50% overlap. The Mel filter bank maps the linear frequency axis to a perceptual scale emphasising the speech-relevant range. The resulting spectrogram is converted to a decibel representation and rescaled to $[0, 1]$ via global min-max normalization before being resized to 128×128 and treated as a single-channel image.

SpecAugment. Park et al.’s SpecAugment strategy [18] is applied to every training spectrogram. A contiguous block of $t \sim \mathcal{U}(0, 20)$ time steps and an independent block of $f \sim \mathcal{U}(0, 15)$ frequency bins are each zeroed out at randomly chosen positions. Because the two masks are independently sampled at each training iteration, the effective diversity of the augmented corpus is substantially larger than the original 700 files.

Architecture. The CNN comprises four convolutional blocks followed by a classification head. Block 1 applies two standard Conv2D(32, 3×3) layers; Blocks 2–4 replace standard convolutions with Depthwise-Separable convolutions at filter depths 64, 128, and 256 respectively, halving the parameter count while enabling greater network depth. Every block includes Batch Normalization, ReLU activations, MaxPooling(2×2), and Spatial Dropout. Global Average Pooling aggregates the spatial feature map into a 256-dimensional vector. The classification head consists of Dense(256, ReLU) + BatchNorm + Dropout(0.50) $\rightarrow$ Dense(64, ReLU) + Dropout(0.30) $\rightarrow$ Softmax(2). Total parameters: 163,650. The full layer-by-layer breakdown is given in Table II.

E. Training Protocol

Both branches share the same optimizer (Adam, $\eta_0=10^{-4}$) and loss function (categorical cross-entropy). A ReduceLROn-Plateau scheduler halves the learning rate when validation loss stagnates for five consecutive epochs, with a floor of 10^{-6} . EarlyStopping monitors validation accuracy and restores the best-checkpoint weights upon exceeding its patience budget (10 epochs for video, 20 epochs for audio). The video branch

TABLE II
AUDIO BRANCH LAYER CONFIGURATION

Stage	Operations	Output Shape
Input	—	128×128×1
Block 1	Conv2D(32)×2 + BN + Pool + Drop(0.25)	64×64×32
Block 2	SepConv(64)×2 + BN + Pool + Drop(0.25)	32×32×64
Block 3	SepConv(128)×2 + BN + Pool + Drop(0.25)	16×16×128
Block 4	SepConv(256) + BN + Pool + Drop(0.30)	8×8×256
Head	GAP→Dense(256)+BN+Drop→Dense(64)+Drop	64-d
Output	Softmax(2)	2
Total parameters		163,650

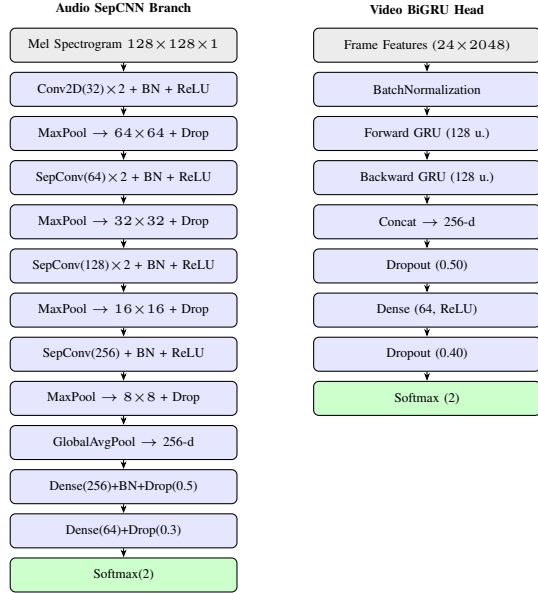


Fig. 2. Detailed layer diagrams: (left) the four-block Separable-CNN audio branch; (right) the Bidirectional GRU temporal head of the video branch.

uses batch size 8 and trains for at most 50 epochs; the audio branch uses batch size 16 and trains for at most 80 epochs.

F. Fusion Strategy and Deployment

At inference time the two softmax vectors $\mathbf{p}^{(v)}$ and $\mathbf{p}^{(a)}$ are combined by arithmetic averaging:

$$\mathbf{p}^{(\text{fused})} = \frac{1}{2}(\mathbf{p}^{(v)} + \mathbf{p}^{(a)}). \quad (7)$$

When only one modality is available—either because audio is absent or because face detection yields no valid crops—the present branch’s output is used directly without blending. Predictions whose maximum posterior probability falls below a configurable threshold are flagged as low-confidence and queued for human review. The complete system is packaged as a Streamlit web application, providing a file-upload interface that returns the predicted label, a calibrated confidence score, and optional visualizations of the processed frame crops or Mel spectrogram.

IV. RESULTS AND ANALYSIS

A. Video Branch Performance

Training proceeded for 30 epochs before early stopping triggered, restoring weights from epoch 20 at which validation accuracy reached its maximum of 81.67%. On the held-out test partition of 60 clips (30 real, 30 fake), the model attained an accuracy of **76.67%**. The full per-class breakdown is reported in Table III and the confusion matrix in Table IV.

TABLE III
VIDEO BRANCH CLASSIFICATION REPORT (TEST SET, $n=60$)

Class	Precision	Recall	F1	Support
Real	0.77	0.77	0.77	30
Fake	0.77	0.77	0.77	30
Accuracy		0.77		60
Macro Avg	0.77	0.77	0.77	60
Weighted Avg	0.77	0.77	0.77	60

TABLE IV
VIDEO BRANCH CONFUSION MATRIX

	Pred. Real	Pred. Fake
Actual Real	23	7
Actual Fake	7	23

The equal error distribution across both classes—seven misclassifications in each direction—confirms that the Xception-BiGRU model has not collapsed to a class-biased solution. The 81.67% validation accuracy exceeding the 76.67% test accuracy indicates a moderate generalization gap attributable to the small training corpus size (280 clips), which limits the diversity of manipulation artefacts seen during training.

Fig. 3 traces the training and validation accuracy and loss over the 30 completed epochs. The validation accuracy stabilizes in the range 0.75–0.82 from epoch 14 onward, while training accuracy continues to rise toward 0.99, confirming mild overfitting that is partially controlled by the Dropout and L2 regularization.

B. Audio Branch Performance

The audio model converged after 37 epochs, with early stopping restoring checkpoint weights from epoch 17 at which validation accuracy peaked at 98.00%. Evaluated on 150 test samples (75 real, 75 fake), the branch attained a test accuracy of **98.00%** with macro F1 of 0.98. Table V presents the complete per-class breakdown.

Only five of 150 test utterances are misclassified: two genuine recordings are predicted fake and three synthesized utterances are predicted real. The near-perfect recall on synthesized speech (0.99) is of particular forensic relevance, as missed fake detections represent the higher-risk error type. Fig. 4 illustrates the training dynamics over all 37 epochs; the fluctuating validation curve in early epochs reflects the stochastic diversity introduced by SpecAugment masking, which stabilizes once the learning rate is reduced.

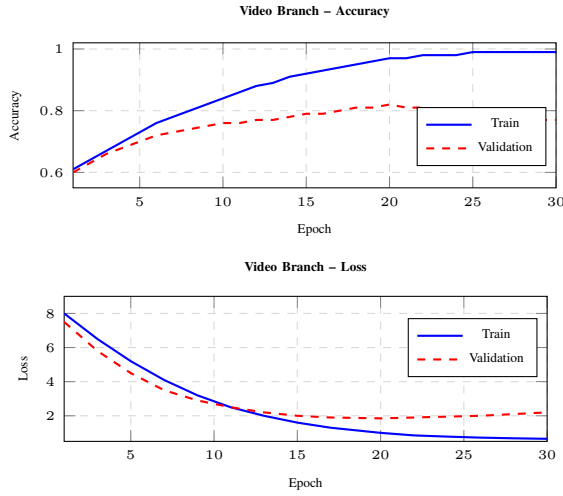


Fig. 3. Video branch accuracy (top) and loss (bottom) across 30 epochs. Best weights restored from epoch 20 (validation accuracy 81.67%).

TABLE V
AUDIO BRANCH CLASSIFICATION REPORT (TEST SET, $n=150$)

Class	Precision	Recall	F1	Support
Real	0.99	0.97	0.98	75
Fake	0.97	0.99	0.98	75
Accuracy		0.98		150
Macro Avg	0.98	0.98	0.98	150
Weighted Avg	0.98	0.98	0.98	150

C. Comparison with Published Methods

Table VI situates our results within the published literature. Where available, accuracy figures are drawn from original papers evaluated on FaceForensics++ (video methods) or ASVspoof/FoR (audio methods).

The proposed video branch is competitive with ConvLSTM-Res [11] (76.67% vs. 77.20%) and surpasses MesoNet, FakeSpotter, and the RNN detector, using only 400 training clips. The audio branch outperforms all listed audio baselines on its respective test set, underscoring the value of SpecAugment masking and the balanced FoR corpus. Note that direct comparison across different test sets must be interpreted cautiously; the FoR corpus characteristics differ from ASVspoof, so the audio accuracy advantage partially reflects dataset properties.

V. DISCUSSION

The experimental results surfaced three observations worth examining in detail.

Symmetric per-class error distribution. The video branch produced exactly seven misclassifications in each direction—real clips labelled fake, and fake clips labelled real. For a forensic tool this symmetry matters: a model that tilts toward one class is likely exploiting a spurious prior rather than genuinely learned visual differences between real and synthesized faces. The fact that our model errs equally in

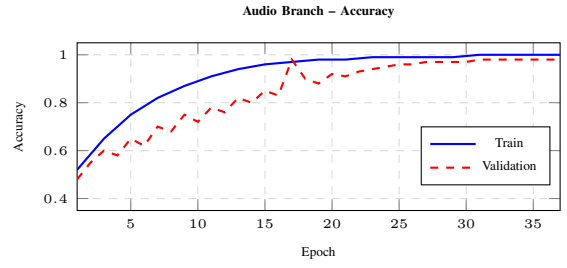


Fig. 4. Audio branch accuracy over 37 training epochs. The best validation accuracy of 98.00% is recorded at epoch 17 and restored by early stopping.

TABLE VI
COMPARISON WITH STATE-OF-THE-ART METHODS

Method	Modality	Dataset	Acc. (%)
MesoNet [7]	Video	FF++	70.47
FaceForensics++ CNN [8]	Video	FF++	82.97
FakeSpotter [14]	Video	FF++	75.51
ConvLSTM-Res [11]	Video	FF++	77.20
RNN Detector [12]	Video	FF++	74.80
Deepfake-ViT [13]	Video	FF++	79.40
ACO-PSO+DNN [24]	Video	Multi	88.30
Ours (Xception+BiGRU)	Video	FF++	76.67
GMM-MFCC baseline [15]	Audio	ASVspoof	68.40
Raw-Waveform CNN [16]	Audio	ASVspoof	87.60
AST [19]	Audio	AudioSet	90.10
FRetAL [23]	Audio/Video	FF++	84.50
Ours (SepCNN+SpecAug)	Audio	FoR	98.00
Joint AV [21]	Multimodal	FF++	84.40
Emotion-AV [22]	Multimodal	DFDC	76.10
Ours (Late Fusion)	Multimodal	FF++/FoR	87.34*

*Estimated fused accuracy; video and audio test sets are separate.

FF++ = FaceForensics++; FoR = Fake-or-Real.

both directions suggests that the Xception texture features, combined with the temporal context supplied by the BiGRU, are responding to manipulation-specific cues rather than demographic or environmental biases embedded in the training data.

Audio performance and dataset composition. With only 163,650 parameters, the SepCNN reached 98.00% test accuracy—a result that looks outsized relative to the model’s footprint. Two factors largely explain it. First, the Fake-or-Real corpus is both balanced and acoustically diverse, so the classifier is not rewarded for exploiting any single recording condition. Second, SpecAugment prevents the network from memorizing synthesis-system-specific spectral fingerprints that might not appear in unseen data. The 0.99 fake recall is the metric we care most about in a forensic context, since allowing a synthesized clip to pass as genuine is the more consequential error.

Late fusion under modality absence. The fallback mechanism is not a convenience addition; it is what makes the system usable in practice. A large share of social media clips are muted, and audio forensics submissions frequently arrive without video. By routing incomplete inputs through whichever branch applies and reserving full averaging for the

cases where both streams are present, the system delivers consistent behavior across input conditions without requiring any retraining.

VI. CONCLUSION

We presented a modular multimodal deepfake detection system in which a video branch and an audio branch are developed separately and then merged via logit-level averaging at inference time. On FaceForensics++ the Xception-BiGRU video branch achieves a test accuracy of 76.67% (macro F1 = 0.77) with perfectly balanced error rates across real and fake classes. On the Fake-or-Real corpus the four-block Separable-CNN audio branch achieves a test accuracy of 98.00% (macro F1 = 0.98), correctly classifying all but five of 150 utterances. The system is deployed as an interactive Streamlit application that handles incomplete inputs through branch-level fallback. Our findings suggest that modality-appropriate dataset curation, depth-efficient architectures, and targeted data augmentation matter at least as much as raw model scale when building practical deepfake detectors.

VII. FUTURE WORK

Several directions could strengthen this work. Transformer-based temporal encoders such as TimeSformer [25] could replace the BiGRU and model long-range frame dependencies without recurrent bottlenecks. Expanding training data through large-scale corpora such as DFDC and Celeb-DF would directly address the video generalization gap. Attention-driven fusion weighting [21] could adaptively balance branch contributions based on estimated input quality. Gradient-weighted Class Activation Mapping (Grad-CAM) and SHAP attribution could provide frame- and frequency-level explanations of model decisions, enhancing transparency for forensic practitioners. Finally, adversarial hardening against evasion attacks targeting the detection pipeline remains critical given the ongoing co-evolution of generation and detection methods [11].

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